

Clinical Study Summary Report (CSSR)

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1. Administrative & Study Identification

Full Study Title: Effectiveness of ABRYSV0® Maternal Respiratory Syncytial Virus (RSV) Vaccine Against RSV in Infants in Western Pennsylvania

Short Title/Acronym: CASSATT

Protocol Number: C3671074

NCT Number: NCT06813872

Sponsor Name: Pfizer

Investigational Product: ABRYSV0® (Bivalent RSV prefusion F protein-based vaccine / RSVpreF)

Phase of Development: Phase IV / Observational (Post-licensure Real-World Effectiveness)

Medical Monitor: Pfizer Clinical Operations Lead

2. Study Synopsis & Rationale

2.1 Study Objective Primary Objective: To evaluate the vaccine effectiveness (VE) of ABRYSV0 vaccination during pregnancy against RSV-associated lower respiratory tract disease (LRTD) outcomes (such as hospitalization) in infants in a real-world population in Western Pennsylvania using a Test Negative Design (TND). **Secondary Objectives:** To evaluate VE against severe RSV-associated LRTD leading to hospitalization among infants aged 0 to ≤ 6 months.

2.2 Background & Rationale Globally, respiratory syncytial virus (RSV) is the leading cause of lower respiratory tract disease (LRTD) in infants and children. Pfizer has developed ABRYSV0, which is approved in the United States and recommended for the active immunization of pregnant individuals from 32\ 0/7\$ to 36\ 6/7\$ weeks' gestational age to prevent LRTD in infants from birth through 6 months of age. To generate critical evidence to support vaccine policy and implementation, this study collaborates with the University of Pittsburgh to assess real-world VE over multiple RSV seasons.

3. Study Design & Methodology

3.1 Design Framework Study Type: Observational (Retrospective/ Prospective Electronic Health Record extraction) **Control Type:** Active Comparator (Test-negative controls: infants meeting the clinical outcome definition but with a laboratory-confirmed negative RSV test) **Blinding:** Open-label / Non-blinded (Real-world data) **Randomization:** N/A (Test Negative Design)

3.2 Planned Enrollment & Duration Target **\$N\$:** Determined by the total number of eligible hospitalizations at UPMC facilities across the 2023-2027 RSV seasons **Number of Sites:** UPMC facilities in Western Pennsylvania, United States **Estimated Study Start:** 24-Feb-2025 **Estimated Primary Completion:** 31-Jul-2027

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Infants ≤ 6 months (≤ 180 days) of age on the index date. 2. Hospitalized at a UPMC facility during the RSV season (approx. 01 October to 15 April) with acute respiratory tract illness symptoms meeting the protocol-defined clinical case. 3. Infants born at ≥ 32 weeks of gestation to individuals expected to reach the indicated ABRYSVO vaccination window ($32\ 0/7$ to $36\ 6/7$ weeks' gestation) during the local vaccination season. 4. Infant's EHR successfully links to the maternal EHR, with known RSV testing results from a specimen collected 10 days prior to hospital admission through 3 days after.

4.2 Exclusion Criteria 1. Infants who received any licensed or investigational RSV preventive product (e.g., RSV monoclonal antibodies, RSV vaccine) since birth. 2. Born to a birth parent for whom ABRYSVO vaccination status cannot be verified in the maternal EHR or Pennsylvania Immunization Electronic Registry System (PIERS) system. 3. Infants born to an individual vaccinated with ABRYSVO prior to the current pregnancy, vaccinated with more than one dose, or vaccinated with any other licensed/investigational RSV vaccine during pregnancy.

5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: Single maternal dose of ABRYSVO. **Route of Administration:** Intramuscular (administered to the birth parent during pregnancy). **Duration of Treatment:** Single administration during the $32\ 0/7$ to $36\ 6/7$ weeks' gestation window. The follow-up observation window applies to the infant from birth through 6 months of age.

5.2 Concomitant & Prohibited Therapy Permitted: Standard of care therapies for pregnant individuals and pediatric respiratory hospitalizations. **Prohibited:** Receipt of RSV monoclonal antibodies (e.g., nirsevimab) or other RSV vaccines by the infant.

6. Study Timeline & Visit Schedule

(Note: As this is a real-world observational study utilizing EHR data with no active enrollment or direct participant contact,

standard clinical visits are replaced by data extraction milestones.)

Visit ID	Window (Days)	Key Activities/Procedures	
Maternal Exposure	Gestation Wk 32\ 0/7\$ to 36\ 6/7\$	ABRYSVO Administration (Historical/EHR Verification)	
Infant Birth	Day 0	Linkage of Infant EHR to Maternal EHR / PIERS	
Hospitalization (Index)	Birth to \$le 180\$ Days	Admission for acute respiratory tract illness; Collection of RSV testing sample	
Testing Window	-10 to +3 Days from Index	Laboratory confirmation of RSV (Positive for Cases, Negative for Controls)	

7. Outcome Measures (Endpoints)

7.1 Primary Endpoint Definition: Vaccine Effectiveness (VE) of maternal ABRYSVO vaccination against RSV-associated hospitalization in infants. Cases are defined as infants with a positive laboratory-confirmed RSV test; Controls are infants with a laboratory-confirmed negative RSV test. **Measurement Method:** Laboratory testing of respiratory samples collected within 10 days prior to hospital admission through 3 days after admission.

7.2 Secondary & Exploratory Endpoints 1. VE of maternal ABRYSVO against severe RSV-associated LRTD leading to hospitalization among infants.

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Retrospectively extracted from UPMC Electronic Health Records (Standard of Care data). No active primary data or direct contact with participants occurs outside of routine care. **Serious Adverse Events (SAEs):** Captured inherently via hospital admission records in the EHR system. **Data Monitoring Committee (DMC):** Data governance and sharing managed collaboratively between Pfizer and the University of Pittsburgh.

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Test-Negative Design (TND) Population. Infants are considered "exposed" if their birth parent has documented evidence of receiving ABRYSVO \$ge 14\$ days before the infant's birth. **Sample Size Justification:** Determined by the available real-world population hospitalized at UPMC facilities across four RSV seasons (2023-2024 through 2026-2027). **Handling of Missing Data:** Infants are strictly excluded from the VE analyses if maternal ABRYSVO vaccination status cannot be definitively verified via EHR or the PIERS system.

10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study will be conducted in accordance with Good Pharmacoepidemiology Practices (GPP) and local privacy and data protection laws (e.g., HIPAA). **Monitoring Plan:** Systematic querying of the UPMC EHR and Pennsylvania Immunization Electronic Registry System (PIERS). **Audit Trail:** Secure, de-identified database maintenance and verifiable EHR linkage algorithms applied by the University of Pittsburgh.

11. Study Identifiers & Governance

Primary ID: C3671074 **Registry Number:** NCT06813872 **Sponsor/Lead Organization:** Pfizer **Study Title:** CASSATT **Official Regulatory Title:** Effectiveness of ABRYSV0® Maternal Respiratory Syncytial Virus (RSV) Vaccine Against RSV in Infants in Western Pennsylvania

12. Clinical Framework (RSV Specific)

Primary Condition: Respiratory Syncytial Virus (RSV) associated Lower Respiratory Tract Disease (LRTD) **Diagnosis Threshold:** Positive laboratory-confirmed RSV test from a respiratory sample collected ≤ 10 days prior to or ≤ 3 days after hospital admission **Medical Methodology:** Maternal Immunization / Preventive Vaccine **Optimization Target:** Prevention of infant RSV-LRTD hospitalization

13. Study Procedures & Methodology

Management Pathway: Standard of Care (SOC) hospital admission and diagnostic RSV testing **Education Protocol (HCP):** N/A (Observational Real-World Study) **Education Protocol (Patient):** N/A (No direct participant contact) **Quality Audit Mechanism:** Automated EHR data linkage mapping (infant to maternal records) and PIERS registry vaccination verification

14. Observation & Follow-Up Schedule

Total Duration: Birth through ≤ 6 months (≤ 180 days) of age **Onsite Visit Cadence:** N/A (Triggered by real-world hospital admission during local RSV seasons approx. Oct 1 to Apr 15) **Remote Monitoring Frequency:** Continuous/periodic retrospective EHR extraction **Checklist Review Interval:** N/A

15. Sign-off & Approval

Role	Name	Signature	Date						
Principal Investigator	[University of Pittsburgh PI]	_____	[DD-MMM-YYYY]						
Clinical Operations Lead	[Pfizer Representative]	_____	[DD-MMM-YYYY]						
Lead Statistician	[Lead Statistician Name]	_____	[DD-MMM-YYYY]						